

Zakaria Khan

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SUMMARY

Undergraduate in chemical physics. Interested in novel, broad expressions of science.

EXPERIENCE

U.S. Dept of State, Universiti Malaya

Aug 2025 – pres

Fulbright Scholar

Kuala Lumpur, Wilayah Persekutuan, MY

- Conducting Monte Carlo simulations and density functional theory (DFT) studies to analyze electron transport properties of novel quantum materials, including topological insulators, 2D materials, and hybrid perovskites.
- Developing hybrid simulation pipelines that integrate DFT's deterministic quantum mechanical framework with Monte Carlo's probabilistic sampling approach.
- Leading research to investigate how environmental factors (e.g., temperature, electric, and magnetic fields) affect carrier mobility and electron scattering in novel quantum materials.

Deutsche Forschungsgemeinschaft Young Scientist

Jun 2025 – July 2025

Department of Energy Laboratory Intern

Lindau, Bavaria, DE

- Participated in talks on quantum information and machine learning applications, deepening knowledge in emerging technologies relevant to next-generation computing platforms and material sciences.
- Contributed to discussions on scientific diplomacy and global scientific challenges, representing the U.S. academic community in an international forum.

Wayne State University

Jun 2021 – Apr 2025

Research Assistant, Barry M Goldwater Scholar

Detroit, MI, USA

- Investigated attosecond spectroscopies, contributing to the development of ultrafast optical techniques for probing electron dynamics on femtosecond and attosecond timescales.
- Designed materials inspired by THz spectroscopies, focusing on materials with advanced electronic properties for potential applications in next-gen semiconductors and photonics.
- Developed coding techniques to analyze experimental data and optimize measurement protocols, advancing the lab's capabilities in time-resolved spectroscopy.

Los Alamos National Laboratory

May 2024 – Aug 2024

Department of Energy Laboratory Intern

Los Alamos, NM, USA

- Worked on the simulation and analysis of coagulation microphysics in particle dynamics, supporting ongoing research in fluid dynamics and particle interactions at the atomic and molecular levels.
- Applied computational methods to model particle behavior under various conditions, contributing to understanding of fundamental processes in material fabrication and nanotechnology.
- Presented research findings to senior scientists, contributed to a major report.

EDUCATION

Wayne State University

May, 2025

B.S. Chemistry, Physics

Detroit, MI, USA

SKILLS, INTERESTS

- **Skills:** Spectrophotometry, optical instrumentation, quantum optics, experimental quantum information

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